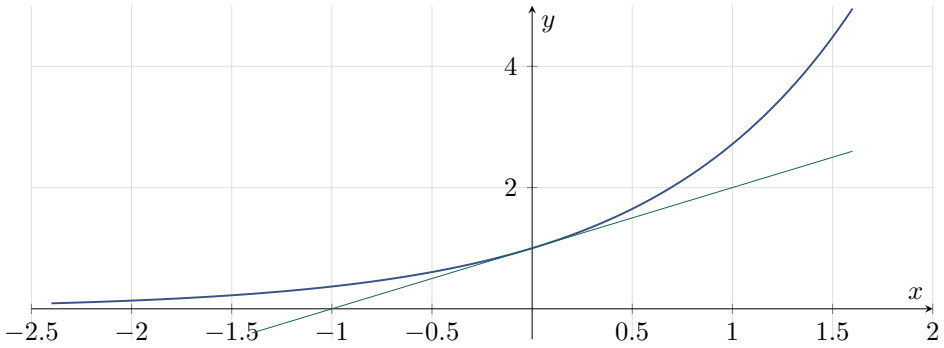
9.1 Defining a new function

Definition and results

The exponential function is the only ferivable function on $\mathbb{R}$ verifier $f\prime = f$ and $f(0) = 1$.
Its existence and uniqueness are admitted. expthen e^x with $e = \exp(1) \approx 2{,}71828$.
The number e is a constant; x is the variable. The exponential is strictly positive.

Status and justification. Existence and uniqueness admitted in accordance with the programme. Deduced, after the algebraic properties allowed below, from $\exp(x) = \exp(\frac{x}{2})^2$ and $\exp(x)\exp(-x) = 1$.

Example solved. $\exp(0) = 1$, $\exp(1) = e$, exp ′(0) = 1. The tangent in 0 is therefore $y = 1 + x$. The Rating e^x extends power properties to any real exhibitor.



e^x (blue) and its tangent 1 + x in 0 (green).

Point of vigilance

e^0 = 1, not 0. The function never takes the value 0, even for negative xes.

Immediate application. What is the equation of tangent to e^x One?

9.2 Algebraic properties and comparisons

Definition and results

For all real x,y, $e^{x+y} = e^x e^y$, $e^{-x} = \frac{1}{e^x}$, $e^{x-y} = \frac{e^x}{e^y}$. For integer n, ($e^x)^n = e^{nx}$. The function is strictly increasing on $\mathbb{R}$, therefore $e^u = e^v \iff u = v$ and $e^u < e^v \iff u < v$.

Status and justification. The relationship of addition is accepted in the common path; it can be deepened through uniqueness. Positivity and $\exp' = \exp > 0$ give the strict growth. justifies equivalences on equations and inequities.

Example solved. $e^{2x}e^{1-x} = e^{x+1}$. $e^{3x-2} = e^{x+4}$ equals 3 $x - 2 = x$ +4, so x=3. $e^{2x} \leq 1 = e^0$ is equivalent to $x \leq 0$.

Point of vigilance

e^{x+y} is a product, never in general $e^x + e^y$. We don't solve e^x = 2 with logarithm Not studied first.

Immediate application. Simplify $\frac{e^{3x}}{e^{x-1}}$; solve $e^{x+2} > e^{2x}$.

9.3 Derive and Study

Definition and results

For real, $t \mapsto e^{at}$ is derivable and its derivative is ae^{at}. For $e^{at+b} = e^b e^{at}$the derivative is ae^{at+b}. Product rules and quotients apply with already known derivatives.
It often simplifies the study of the derivative sign.

Status and justification. The formula of derivation of e^{at} is allowed; the case with b ensues from it exiting the constant factor e^b. The general compound bypass is not necessary here.

Example solved. For $f(x) = (x+1)e^{-x}$, $f'(x) = e^{-x} - (x+1)e^{-x} = -xe^{-x}$. Like $e^{-x} > 0$, the derivative has the sign of $-x$: total maximum 1 in 0.

Point of vigilance

Do not forget the factor has nor the negative sign in e^{-x}. The exponential factor is positive, not necessarily increasing.

Immediate application. Dirty $g(t) = 3e^{-2t}$ and study its sense of variation.

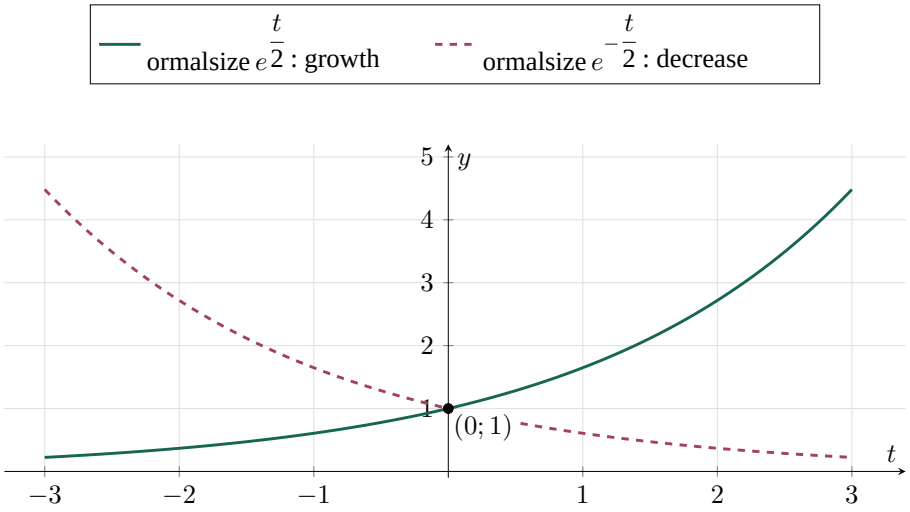
9.4. From discrete to continuous

Definition and results

A model $Q(t) = Q_0 e^{kt}$, $Q_0 > 0$, represents growth if $k > 0$, a decrease if $k < 0$, a consistency if $k = 0$. For a fixed pitch h, $\dfrac{Q(t+h)}{Q(t)} = e^{kh}$. Values at moments integer form a geometric sequence of common ratio e^k. The unit of k is the inverse of the unit of time.

Status and justification. The report is based on: $e^{k(t+h)} = e^{kt}e^{kh}$. The direction of variation is derived from $Q\prime(t) = kQ_0e^{kt}$ and positivity of the remaining factor.

Example solved. $Q(t) = 100e^{-0,2t}$T in hours: $Q(0) = 100$, $\dfrac{Q(t+1)}{Q(t)} = e^{-0,2}$. A simulation can look for the first integer number of hours where Q passes under 50; the exact equation is not solved by logarithm in this chapter.



The curves corresponding to $k = \dfrac{1}{2}$ and $k = -\dfrac{1}{2}$ are symmetrical in relation to the vertical axis. trait styles also distinguish them without color.

Point of vigilance

The relative rate per unit of discrete time is $e^k - 1$, not exactly k; they are close Only for k small.

Immediate application. For $Q(t) = 5e^{0,3t}$, express $Q(n)$ and $\dfrac{Q(t+2)}{Q(t)}$.

9.5 Digitally Build and Demonstrate Inequality

Definition and results

To approach the solution $y' = y,\ y(0) = 1$ with one step h, Euler replaces the increase by $hy : y_{k+1} = (1+h)y_k$. On [0; 1] with $h = \frac{1}{n}$, the approximation of e is $\left(1+\frac{1}{n}\right)^n$. This construction is experimental. $e^x \geq 1 + x$ obtained by function study.

Status and justification. For $g(x) = e^x - 1 - x,\ g'(x) = e^x - 1$, negative if x<0 and positive if x>0. The minimum is $g(0) = 0$, which proves inequality; equality only in 0. Convergence o Euler is not demonstrated here.

Example solved. With n=2, Euler gives $y_0 = 1, y_1 = 1{,}5, y_2 = 2{,}25$. With n=10, $(1\ ,1)^{10} \approx 2{,}59374$. We compare to e; we do not confuse approximate value with function definition.

Point of vigilance

A graph showing a curve above a straight line does not prove inequality over all the real.

Immediate application. Show that $e^{2x} \geq 1 + 2x$ for all real x.